

SmartClinic 2.0 – Appointment & Patient Management System

PART 1 – System Study

The current clinic system is paper-based and relies on manual processes.

Problems Identified: 1. Appointment delays 2. Miscommunication during rescheduling 3. No patient history tracking 4. Loss of paper records 5. No automated reminders 6. Time-consuming manual work

PART 2 – User Objectives

- Reduce patient waiting time • Automate appointment reminders • Improve appointment accuracy • Avoid double-booking • Improve doctor scheduling • Easy access to patient history

PART 3 – User Requirements

Functional Requirements

- Book appointments • Reschedule appointments • Cancel appointments • Maintain patient records • Doctor availability calendar • Send SMS/Email reminders • Login authentication • Admin dashboard • Search patient history • Generate reports

Non-Functional Requirements

- System loads within 2 seconds • Secure data storage • Mobile-friendly UI • High availability • User-friendly interface • Reliable notification delivery

PART 4 – Software Requirement Specification (SRS)

Purpose: To digitize clinic operations. Scope: Appointment & patient management. Inputs: Patient details, appointment details. Outputs: Confirmation, reminders, reports. Constraints: Clinic hours, authentication required.

PART 5 – Feasibility Study

Technical: Feasible using React, Node, Database. Economic: Low development cost, high benefit. Operational: Staff training required. Conclusion: Feasible

PART 6 – Time & Resource Estimation

UI Design – 1 Week Backend APIs – 1.5 Weeks Database – 0.5 Week Notifications – 0.5 Week Testing & Deployment – 1 Week Team: 2 Interns + 1 Lead

PART 7 – Workflow

Patient → Appointment Booking → Confirmation → Reminder → Consultation → Follow-up